AN INDEPENDENT EXACT VERIFICATION OF THE VARIATIONAL STEP BEHIND THE 2026 BOUNDS $H_1 \le 240$ AND $H_1 \le 236$, WITH A REMARK ON THE CAPS OF A 212 DRAFT

JERRY KIM

ABSTRACT. Several 2026 preprints claim improvements of the bound $H_1 \le 246$ of Polymath8b for $H_1 = \liminf(p_{n+1} - p_n)$. Each combines new equidistribution estimates for primes in arithmetic progressions with a *variational step*: a test function F on a capped region of $\mathbb{R}^k$ whose sieve ratio exceeds a threshold. We recompute the variational step of three of these claims in exact rational arithmetic, with independent programs, and do not address their analytic input. (i) For the support and parameters of Stadlmann ($k = 49$, claimed $H_1 \le 240$) we certify the capped ratio $4.0120389\ldots > 4$ with a test function of our own, and $4.0124198\ldots$ after optimising it on the capped region; two independent programs agree to the last digit. (ii) For the support shared by Kintali and by Song and Yue ($k = 48$, claimed $H_1 \le 236$) we recompute the published witness of Song and Yue: its cap-free mass and its capped numerator equal the paper's reference integers (655 and 715 digits) exactly, so their criterion $48J/I_0 > 2001/2000$ holds, with $48J/I_0 = 1.000664921591\ldots$; the authors list the regeneration of the capped numerator as pending. The same computation reproduces Kintali's claimed capped and cap-free values to every printed digit, and a test function of our own gives $4.0060814\ldots > 4$ on the same support. The capped numerator has so far been recomputed by one program; a second is running. (iii) For the draft of Charton et al. ($k = 45$, claimed $H_1 \le 212$) the criterion holds for the caps of its Table 3 ($4.0027351\ldots > 4$, two programs), whereas for the different caps stated in its Lemma 9.1 no polynomial in an 846-dimensional space of the size the authors use reaches 3.99 (ball arithmetic). This locates an inconsistency in the draft; it is not a refutation. None of this verifies the new equidistribution estimates, on which every one of these bounds rests, and none of it bears on the twin prime conjecture, which remains open.

1. INTRODUCTION

The reduction. For an admissible k-tuple $\mathcal{H} = \{h_1 < \cdots < h_k\}$, let DHL$[k, 2]$ denote the statement that infinitely many translates $n + \mathcal{H}$ contain at least two primes. If DHL$[k, 2]$ holds and $\mathcal{H}$ is an admissible k-tuple of diameter D, then $H_1 \le D$. Maynard's method [4] proves DHL$[k, 2]$ from a level of distribution and a symmetric test function F whose ratio $kJ(F)/I(F)$ exceeds a threshold; Polymath8b [5] enlarged the region on which F may live and obtained $H_1 \le 246$ with $k = 50$. The 2026 preprints listed below enlarge the region further, to *capped* supports, and pay for this with new equidistribution estimates for the moduli that arise.

The claims examined.

claim	k	source	what we check
$H_1 \le 240$	49	Stadlmann [7]	variational criterion
$H_1 \le 236$	48	Kintali [3]; Song–Yue [6]	variational criterion; published witness
$H_1 \le 212$	45	Charton et al. [1]	variational criterion; caps

Date: Version 0.1 (draft), 13 November 2026.

Prepared with the assistance of Claude (Anthropic). All programs, certificates and Lean sources are in the repository https://github.com/jerryckim16/twin-primes; a living companion paper with full details is at https://jerryckim16.github.io/twin-primes/.

Each claim has two parts: an *analytic* part (an equidistribution theorem for the relevant family of moduli) and a *variational* part (a function F on the capped support with ratio above the threshold). We check only the variational part, and the combinatorial fact that an admissible tuple of the stated diameter exists. Throughout we label statements COMPUTED (exact certificate and a script that reproduces it), PROVED HERE, LEAN-CHECKED, LITERATURE (read in the source), or CLAIMED (asserted in a preprint and not verified by us).

Results in brief. Ratios in the Polymath scaling, where the threshold is 4:

claim	variational ratio	status here
240	4.0120389; optimised on T: 4.0124198	COMPUTED, two programs
236, witness of [6]	$48J/I_0 = 1.00066492 > 2001/2000$	COMPUTED, Thm. 5
236, our own F	4.0060814	COMPUTED, one program
212, Table 3 caps	4.0027351	COMPUTED, two programs
212, Lemma 9.1 caps	< 3.99 for all F in an 846-dim. space	COMPUTED, ball arithm.

What this note does not do. It does not verify the equidistribution estimates of [7], [3], [6] or [1]; all four bounds remain conditional on them, and the established bound is still 246. Song and Yue [6, Remark 4.23] point to a missing hypothesis in one step of the proof of [7, Lemma 5]; we read that step and comment on it in Remark 10, without resolving it. Nothing here concerns the parity barrier or the twin prime conjecture.

2. The variational problem on a capped support

Supports. We use the Polymath scaling $u = 4t$, in which the criterion reads $kJ/I > 4$. Let $\mathcal{R}_k = \{u \in [0,1]^k : \ u_1 + \cdots + u_k \le 1\}$ and write $L\mathcal{R}_k$ for its dilate. The one-band supports of [7] (LITERATURE: Definitions 1 and 5 and Proposition 1 of [7]) are described by a length L, a marginal radius τ, a threshold g and caps $C_1 \le C_2 \le \dots$; with $I(u) = \{i : u_i > g\}$, $m(u) = |I(u)|$ and $s(u) = \sum_{i \in I(u)} u_i$,

$$T = \{u \in L\mathcal{R}_k : \ s(u) \le C_{m(u)}\}.$$

For a symmetric F supported on T,

$$I_T(F) = \int_T F^2, \qquad J_T(F) = \int_{\tau\mathcal{R}_{k-1}} \left(\int F(u', u_k)\, du_k \right)^2 du',$$

and the variational criterion is $kJ_T(F)/I_T(F) > 4$. In this scaling the parameters are as follows (LITERATURE).

	k	(L, τ)	g	caps $C_1, C_2, \dots$
[7]	49	(1.042, 0.982)	0.112	0.6, 0.6, 0.68, 0.68, ...
[3], [6]	48	(1.042, 0.982)	91/1250	1899/2500 for all m
[1], Tab. 3	45	(53/50, 249/250)	41/625	$4B_{1,m}$ (see below)
[1], L. 9.1	45	(53/50, 249/250)	41/625	0.62, 0.62, 0.68, 0.68, ...

For [1], Table 3 lists $B_{1,m} = \frac{1}{5000}(777, 794, 875, 917, 953, 983, 1016, 1042, 1063, 1081)$ for $m \le 10$ and $B_{1,m} = 1081/5000$ for $m \ge 11$, and $C_m = 4B_{1,m}$; Lemma 9.1 states $C_1 = C_2 = 31/50$ and $C_m = 17/25$ for $m \ge 3$. All four cap sets satisfy $C_{m+1} \le C_m + g$ (LEAN-CHECKED for the caps of [1] and the counting lemmas below).

Exact moments. All integrals reduce to moments over simplices.

Lemma 1 (Proved here; elementary)**.** *Let $P_j = \sum_i u_i^j$, $X_1, \dots, X_k$ independent exponential variables of mean 1, and $E_k(b) = \mathbb{E}\prod_j P_j(X)^{b_j}$. Then, with $w(b) = \sum_j j b_j$,*

$$\int_{\mathcal{R}_k} (1-P_1)^A \prod_j P_j^{b_j}\, du = \frac{A!}{(k+A+w(b))!}\, E_k(b), \qquad \sum_b E_k(b)\frac{s^b}{b!} = \left(\sum_c w(c)!\,\frac{s^c}{c!}\right)^k.$$

Proof. The Dirichlet integral gives $\int_{\mathcal{R}_k}(1-P_1)^A u^\alpha\, du = A!\,\alpha!/(k+A+|\alpha|)!$ and $\alpha! = \mathbb{E}X^\alpha$; sum over the monomials of $\prod_j P_j^{b_j}$, all of degree $w(b)$. The generating function follows from independence. □

The fibres of a capped support. Let $R = L - |u'|$ on the fibre over $u' \in \mathbb{R}^{k-1}_{\ge 0}$, and call u' *admissible* if $s(u') \le C_{m(u')}$.

Proposition 2 (Proved here; elementary)**.** *If $C_{m+1} \le C_m + g$ for all m, then the fibre $\{t \ge 0 : (u', t) \in T\}$ is empty when u' is not admissible and equals $[0, U(u')]$ otherwise, where*

$$U = \min\big(R,\ \max(g,\ C_{m(u')+1} - s(u'))\big).$$

Consequently, with $Q(u', x) = \int_0^x P(u', t)\, dt$ for a polynomial P,

$$J_T(P\,1_T) = J(P) - \int_{\substack{|u'|\le\tau \\ u' \text{ not adm.}}} Q(u', R)^2 - \int_{\substack{|u'|\le\tau \\ u' \text{ adm., } U<R}} \big(Q(u', R)^2 - Q(u', U)^2\big),$$

and $I_T(P\,1_T) = I(P) - \int_{L\mathcal{R}_k\setminus T} P^2$, where I, J are the cap-free functionals.

Proof. If $t \le g$ the large set of (u', t) is $I(u')$, so $(u', t) \in T$ if and only if u' is admissible and $t \le R$. If $t > g$ the large set gains the last coordinate and the condition is $s(u') + t \le C_{m(u')+1}$; with $C_{m+1} \le C_m + g$ this forces $s(u') < C_{m(u')}$, so a non-admissible u' has an empty fibre, and for admissible u' the fibre is $[0, \min(R, g)] \cup (g, \min(R, C_{m+1} - s)] = [0, U]$. The marginal of $P\,1_T$ is $Q(u', U)$ on admissible u' and 0 otherwise; subtract from the cap-free marginal $Q(u', R)$. □

By symmetry each correction is a finite signed sum, over the size m of the large set and over $j \ge 0$ further coordinates forced above g (inclusion and exclusion for "all other coordinates are at most g"), of integrals of polynomials over polygons in two radial variables; so every term is an exact rational number. The ranges of m and j that occur are short and were checked in Lean (e.g. $m \le 9$, $j \le 3$ for [7]; $m \le 10$, $j \le 3$ for [3]; $m \le 16$, $j \le 6$ for [1]).

3. Method

Two independent exact evaluators. Given a polynomial P with rational coefficients (we use products of $(L - P_1)^a$ and even power sums), both programs return $I_T(P\,1_T)$ and $J_T(P\,1_T)$ as exact rationals.

(1) `capped_exact.py` computes the cap-free values with Lemma 1 and subtracts the corrections of Proposition 2, piece by piece.
(2) `capped_check.py` was written separately from the definitions only, without access to the first program. It sums directly over the exact large set, with exponential-moment generating functions, and has no cap-free term.

Both were tested independently: a Monte Carlo comparison at $k = 12$ with caps that remove almost half of I and J; exact agreement of `capped_check.py` with brute-force polytope triangulation and symbolic integration for $k = 2, \dots, 5$; and exact agreement of the two programs

on every production input reported as "two programs" below. A value counts as certified when both programs return the same two rationals.

Optimising on the capped support, and ceilings. Both functionals are quadratic forms, $I_T = c^\top G_I c$ and $J_T = c^\top G_J c$, in the coefficients c of P. Running the chain of exact linear maps of `capped_check.py` backwards gives the Gram matrices at about the cost of one evaluation (`capped_gram.py`; PROVED HERE, elementary). We use them in two ways. First, to *find* a good vector (the top generalised eigenvector), which is then rounded and re-evaluated exactly by the evaluators; only that exact value is stated as a certificate. Second, to prove a *ceiling*: if $\sigma G_I - kG_J$ is positive definite, no P in the space reaches the ratio σ. Positive definiteness is certified by recursive Schur complements in ball arithmetic, with the entrywise error of G_J included in the balls. The weights behind G_I and G_J were checked against `capped_check.py` exactly at a uniformly random integer vector, which by the Schwartz–Zippel lemma rules out a discrepancy between the two quadratic forms except with probability below 2^{-31} for each.

4. Results

The support of Stadlmann ($k = 49$, claimed $H_1 \le 240$).

Theorem 3 (COMPUTED; exact, two independent programs)**.** *For the parameters of* [7] *in the table above:*

(1) *for P the cap-free optimiser of degree* 20 *in even power sums (*707 *coefficients),*

$$49 J_T(P1_T)/I_T(P1_T) = 4.0120389035\ldots;$$

(2) *for P^* the top eigenvector of $(49G_J, G_I)$ in the same space, rounded to* 400 *significant bits per coefficient,*

$$49 J_T(P^*1_T)/I_T(P^*1_T) = 4.0124198475\ldots.$$

In both cases the two evaluators agree on I_T and J_T as exact rationals. Without caps the same space gives 4.016176, so the caps cost 0.0041 for the truncated optimiser and 0.0038 for the optimised one: optimising on T regains less than a tenth of the loss. Since every point of T has all coordinates at most 0.6 and $T \subset L\mathcal{R}_{49}$, a dual-weight bound gives the ceiling $\sup_F kJ_T/I_T \le 4.14$ (COMPUTED, interval arithmetic). So the margin of the variational step of [7] lies in (0.0124, 0.14].

Corollary 4 (conditional; the analytic input is CLAIMED)**.** *The criterion of* [7, Prop. 1] *holds for $k = 49$ with Stadlmann's support, caps and parameters. With the* LEAN-CHECKED *admissible* 49*-tuple of diameter* 240*, $H_1 \le 240$ follows from* [7, Prop. 1] *and his equidistribution theorem, which we have not verified.*

The support of Kintali and of Song–Yue ($k = 48$, claimed $H_1 \le 236$). Both preprints use the same support (the table above, called "D182" in [3]) and, by their descriptions, the same public witness of degree 25, originally constructed for $k = 50$ and evaluated at $k = 48$ without re-optimisation. Song and Yue print it as 1295 integer triples (a, α, c), with

$$P(z) = \sum c\,\left(\tfrac{26}{25} - s(z)\right)^a m_\alpha(z), \qquad P_U(t) = P\left(\tfrac{26}{25U}t\right), \quad U = \tfrac{521}{2000},$$

m_α the monomial symmetric polynomials, and they print the reference integers $I_{\mathrm{ref}} = (98!)^2 I_0^*$ and $J_{\mathrm{ref}} = (99!)^2(26!)^2 J^*$. Here I_0 is the *cap-free* mass on $\{s < U\}$, J the capped numerator, and the stars denote integer coordinates $x = 10^4 t$. Their criterion is $48J/I_0 > 2001/2000$, which implies the capped criterion because $I_T \le I_0$ (LITERATURE, [6, §4.12, App. A, App. C]). They state that a fresh evaluation reproduces I_{ref}, while fresh regeneration of J_{ref} from the table "remains incomplete".

We extracted the table from the PDF using font sizes to read the exponents in the partitions. All 1295 rows are present, every α is nondecreasing with even parts, and $\max(a+|\alpha|)=25$, as stated. Since all parts are even, P_U is exactly a polynomial in $L-P_1$ and the even power sums $P_2,\dots,P_{24}$ (1311 terms); the conversion was checked exactly at random rational points.

Theorem 5 (COMPUTED, exact rational arithmetic)**.** *For the witness P_U of* [6] *and the support above:*

(1) $(98!)^2\,10^{192}\,5^{150}\,521^{50}\,I_0 = I_{\mathrm{ref}}$ *exactly (an integer with* 655 *digits);*
(2) $(99!)^2(26!)^2\,10^{196}\,5^{150}\,521^{50}\,J_T = J_{\mathrm{ref}}$ *exactly (an integer with* 715 *digits), where J_T is the capped numerator;*
(3) *hence* $48J_T/I_0 = 1.000664921591\ldots > 2001/2000$*, which is* $4.0026596864\ldots$ *in the Polymath scaling; with the capped denominator* $48J_T/I_T = 4.0030988047\ldots$*, and without caps* $48J_0/I_0 = 4.0036348167\ldots$.

The factors $10^{192}5^{150}521^{50}$ and $10^{196}5^{150}521^{50}$ come from the change to integer coordinates when the polynomial is scaled by 65125^{25} (our reading of [6, Eq. (71)]; $65125 = 25\cdot 2605$), so items (1) and (2) are equalities of integers, not numerical agreement. Item (1) was computed by a program written for this purpose, which works in the monomial basis of the table, integrating each product $m_\alpha m_\beta$ by a sum over partial matchings of parts, and does not use the power-sum conversion. Item (2) was computed by `capped_check.py` on the converted polynomial (56 minutes on four cores). The capped and cap-free values in (3) agree with those claimed in [3], 4.0030988 and 4.0036348, to every printed digit.

Status of item (2). This item and Proposition 6 below are the results of this note that so far rest on one program. The second evaluator, `capped_exact.py`, agrees with it on the cap-free part and is computing the 126 correction pieces from a checkpoint (about 5–10 hours on four cores). This draft will be updated when it finishes. The exact agreement of the first program with the authors' independently produced 715-digit integer is itself strong evidence for both, since a common error would have to produce the same integer.

Proposition 6 (COMPUTED, exact rational arithmetic, one program)**.** *For our own cap-free optimiser of degree* 22 *for this (L,τ) at $k=48$ (*1041 *coefficients, even power sums, rounded to* 400 *significant bits),* $48J_T(P1_T)/I_T(P1_T) = 4.0060814208\ldots > 4$.

Here the caps cost only 0.00033 (the cap-free value is 4.0064115). That our value exceeds the witness's 4.0031 is expected, since the witness was optimised for $k=50$ and transported, while ours is optimised for $k=48$.

Corollary 7 (conditional; the analytic input is CLAIMED)**.** *The variational criteria of* [3] *and* [6] *hold at $k=48$, for the published witness and for an independent test function. With the* LEAN-CHECKED *admissible* 48*-tuple of diameter* 236*, $H_1 \le 236$ follows from their equidistribution input, which we have not verified.*

The draft of Charton et al. ($k=45$, claimed $H_1 \le 212$).

Proposition 8 (COMPUTED; exact, two independent programs)**.** *With the caps of* [1, Table 3] *and P our cap-free optimiser of degree* 21 *(*846 *coefficients, the dimension of the space $\mathcal{B}_{21}$ of* [1, Thm. 11.1]*),* $45J_T(P1_T)/I_T(P1_T) = 4.0027351128\ldots > 4$.

The authors report 4.00438 with their own optimised function (CLAIMED); our value is lower, as expected for a function optimised without caps, and already above 4. The draft, however, also states caps in its Lemma 9.1, namely $C_1 = C_2 = 31/50$ and $C_m = 17/25$ for $m \ge 3$, which are smaller than those of Table 3 for every $m \ge 4$. With these the conclusion changes.

Proposition 9 (Computed, ball arithmetic)**.** *Let T' be the support with the caps of Lemma 9.1 of* [1] *and V the* 846*-dimensional space above. Then $3.99\,G_I - 45\,G_J$ is positive definite on V, so*

$$\frac{45 J_{T'}(P1_{T'})}{I_{T'}(P1_{T'})} < 3.99 \qquad \text{for every nonzero } P \in V.$$

The certificate was obtained twice, at 5000 and 6500 bits with different block splits, and the underlying weights agree exactly with `capped_check.py` at a random integer point. The best P in V gives about 3.98555 (floating point, from the Gram matrices). So within a space of the size the authors use, the caps of Lemma 9.1 cannot give their 4.00438, which must belong to the caps of Table 3. A careful reading of the draft supports this: Table 3 is presented as the parameter set used in both halves of the proof, and the Lemma 9.1 values appear to be left over from an earlier choice. We report this as a located inconsistency for the authors to resolve, not as an error in their result. We have not checked that their space coincides with ours beyond its dimension, and `capped_exact.py` has not yet re-evaluated the random point used for the weights.

5. Remarks

Remark 10 (on [7, Lemma 5]; Literature reading, not a result). Song and Yue [6, Remark 4.23] say that one step in the proof of [7, Lemma 5] lacks a hypothesis. In arXiv v1, Lemma 5 (a Type I estimate) takes $f = \alpha \star \beta$ with α an arbitrary coefficient sequence and β smooth. For γ in a short range above $1/2$ the proof applies Lemma 3 (a Type II estimate) "with the roles of α and β swapped"; Lemma 3 requires its shorter factor to satisfy a Siegel–Walfisz condition, and after the swap that factor is α, for which neither Lemma 5 nor the case of [7, Definition 9] where it is used assumes this. So the step reads as [6] describe it. A natural repair is to check that the sequences α arising in the decomposition have the property; we have not done so, and we do not know whether the conclusion is affected. Theorem 3 concerns only the variational step and is unaffected; the *bound* 240 depends on this input.

Remark 11 (what would change the picture). The variational steps examined here have margins of 0.003 to 0.012 above the threshold. They are therefore robust to rounding but not to a change of support: Proposition 9 shows that a modest change in the caps moves the best attainable ratio below 4. Any revision of the analytic input that shrinks the admissible caps should be re-checked against the variational step, which our programs can do in hours.

6. Reproducibility

All inputs, certificates and logs are in the repository, under `code/` and `code/cap6/`. The evaluators are `code/capped_exact.py` and `code/capped_check.py`; the Gram matrices and ceilings come from `code/capped_gram.py`; the Song–Yue witness and reference integers, as extracted, are `code/cap6/songyue236_witness.json` and `code/cap6/songyue236_appC.json`, with the conversion and comparison scripts `code/songyue_*.py`. Running times above are for four cores and 16 GB of memory. The admissible tuples of diameters 212, 236, 240 and 246 are checked by the Lean 4 kernel (core Lean, no Mathlib, no `sorry`; axioms `propext` and `Quot.sound` only), as are the counting lemmas used in Proposition 2. The companion paper [2] contains the full derivations and the history of each computation.

References

[1] F. Charton, L. Hong, K. Lau, K. Ono, G. Remy, H. C. Siu, A. A. Swaminathan, J. Thorner, Y. Xie, *A new bound for small gaps between primes*, preliminary draft, 3 September 2026. https://primegaps.axiommath.ai/bgp212.pdf

[2] J. Kim, *Toward the twin prime conjecture: certificates, limits of the sieve, and a working programme*, living paper, 2026. https://jerryckim16.github.io/twin-primes/
[3] S. Kintali, *Improved bounded gaps between primes*, preprint, 1 September 2026. https://shivakintali.github.io/papers/Prime-Gaps.pdf
[4] J. Maynard, *Small gaps between primes*, Ann. of Math. 181 (2015), 383–413.
[5] D. H. J. Polymath, *Variants of the Selberg sieve, and bounded intervals containing many primes*, Res. Math. Sci. 1 (2014).
[6] Z. Song, S. Yue, *Bounded gaps between primes: an upper bound of 236*, IACR ePrint 2026/1893, version of 2 September 2026.
[7] J. Stadlmann, *Bounded gaps between primes*, arXiv:2608.31126, 31 August 2026.